

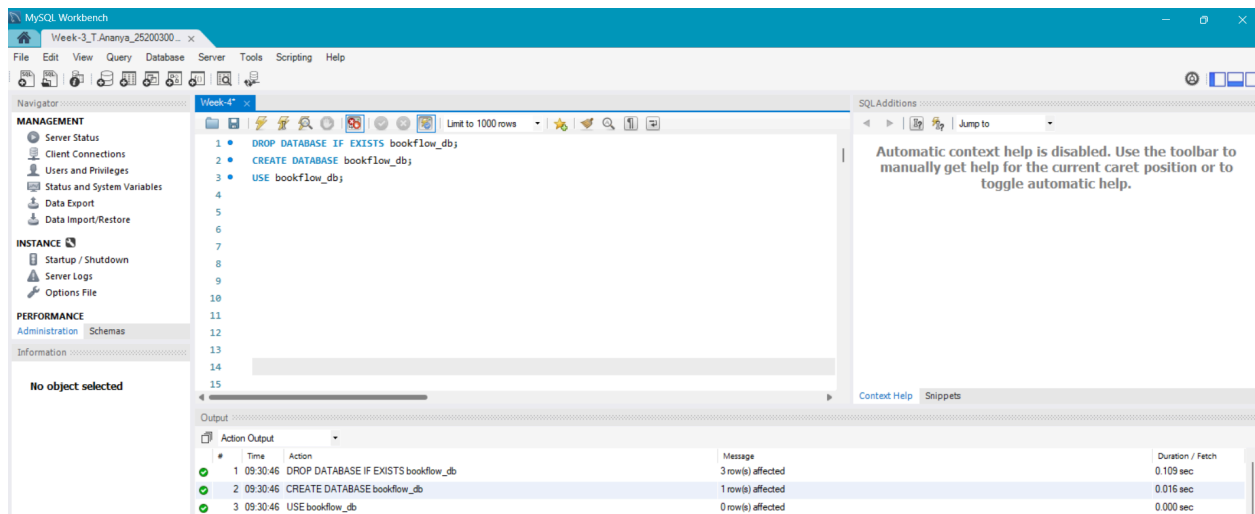
Database System Engineering and Distributed Backend Development

Course Code: 25CS1302E

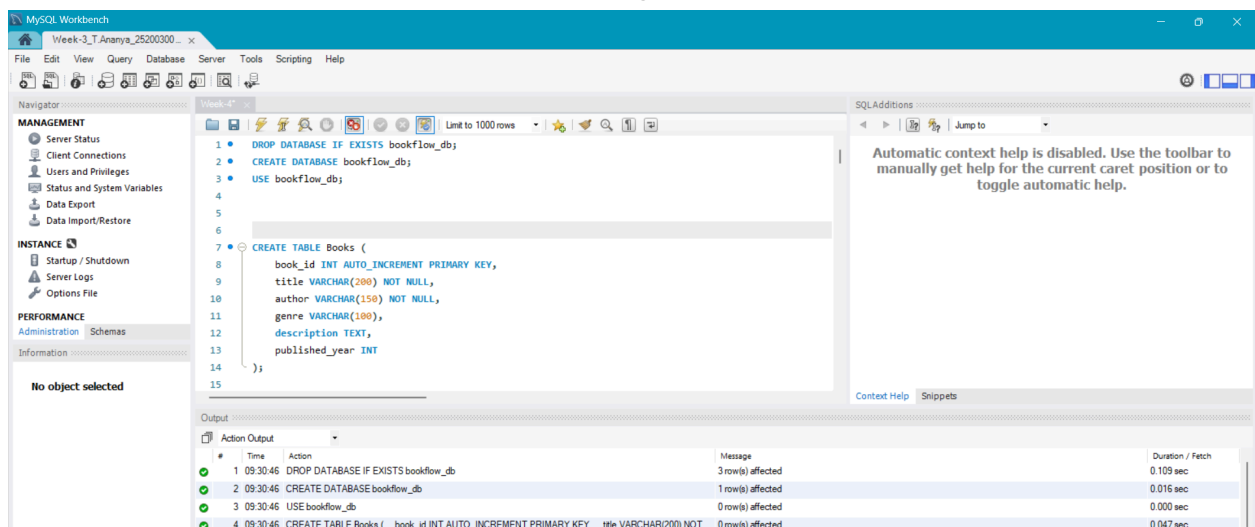
Week -3

T.Ananya
2520030039

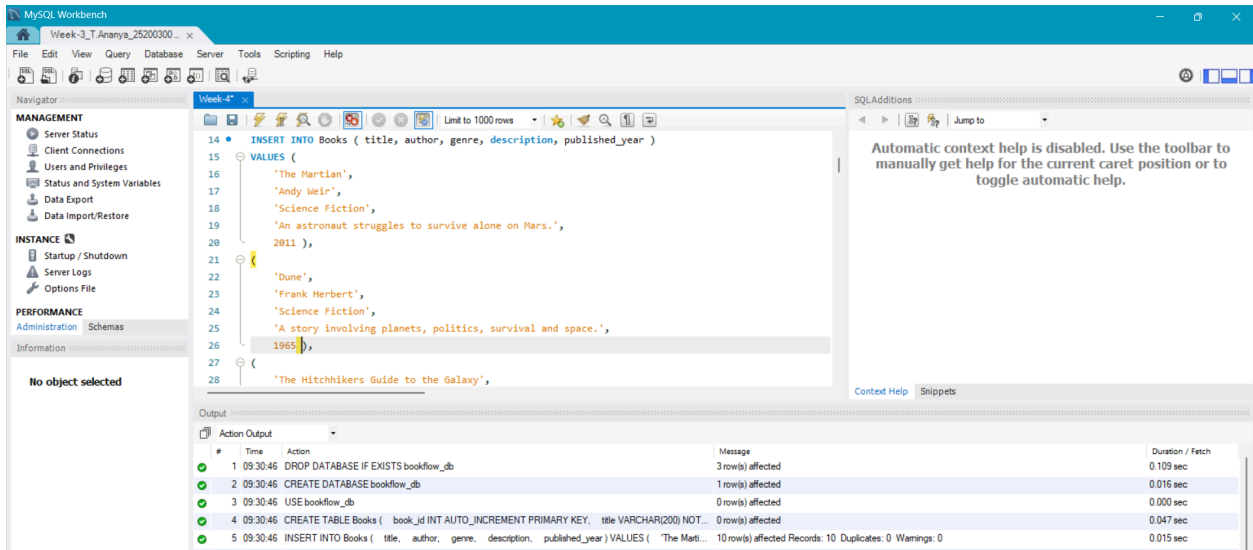
Query-1:



Query-2:



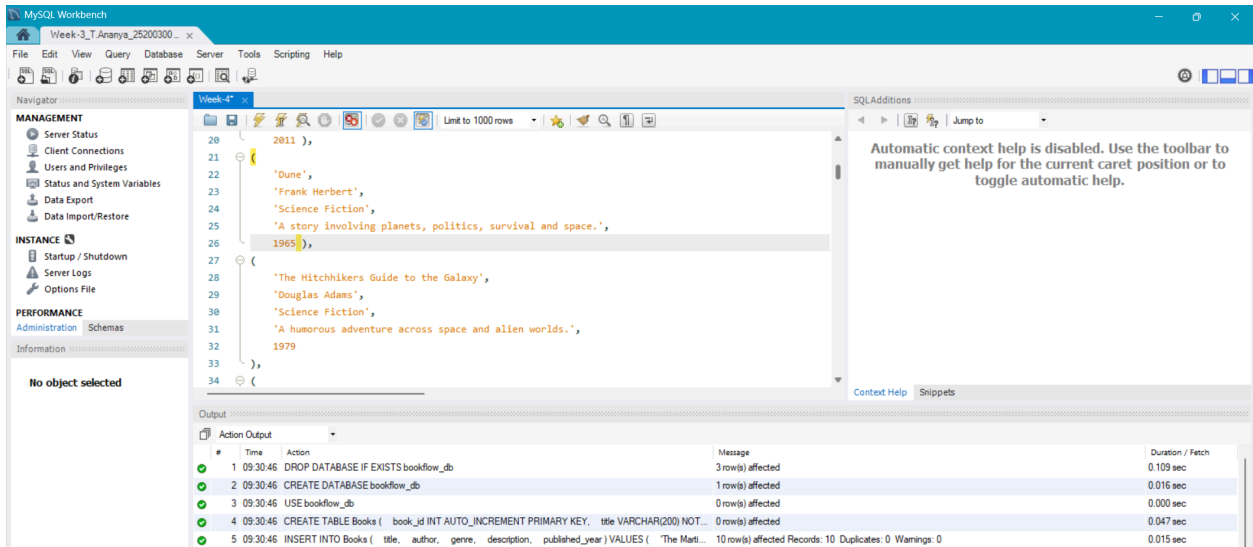
Query-3:



MySQL Workbench interface showing Query-3. The SQL editor contains an INSERT statement into the 'Books' table. The output window shows the execution results.

```
14 INSERT INTO Books ( title, author, genre, description, published_year )
15 VALUES (
16     'The Martian',
17     'Andy Weir',
18     'Science Fiction',
19     'An astronaut struggles to survive alone on Mars.',
20     2011 ),
21
22     'Dune',
23     'Frank Herbert',
24     'Science Fiction',
25     'A story involving planets, politics, survival and space.',
26     1965 ),
27 (
28     'The Hitchhikers Guide to the Galaxy',
```

#	Time	Action	Message	Duration / Fetch
1	09:30:46	DROP DATABASE IF EXISTS bookflow_db	3 row(s) affected	0.109 sec
2	09:30:46	CREATE DATABASE bookflow_db	1 row(s) affected	0.016 sec
3	09:30:46	USE bookflow_db	0 row(s) affected	0.000 sec
4	09:30:46	CREATE TABLE Books (book_id INT AUTO_INCREMENT PRIMARY KEY, title VARCHAR(200) NOT NULL, author VARCHAR(100) NOT NULL, genre VARCHAR(50) NOT NULL, description VARCHAR(255) NOT NULL, published_year INT NOT NULL)	0 row(s) affected	0.047 sec
5	09:30:46	INSERT INTO Books (title, author, genre, description, published_year) VALUES ('The Marti...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0	0.015 sec

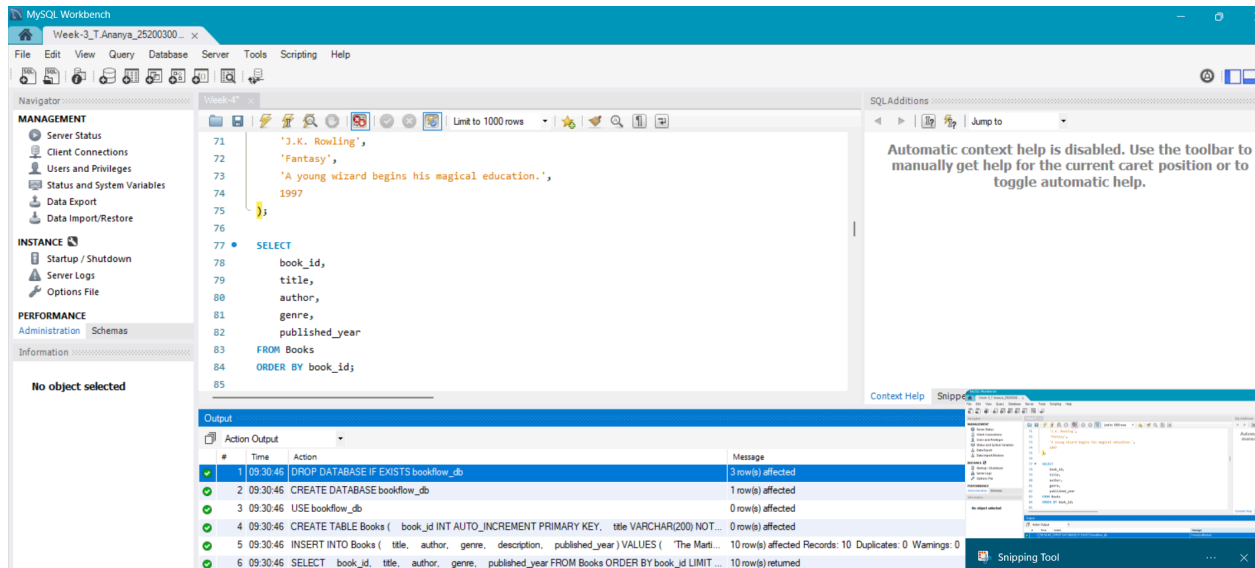


MySQL Workbench interface showing Query-4. The SQL editor contains an INSERT statement into the 'Books' table. The output window shows the execution results.

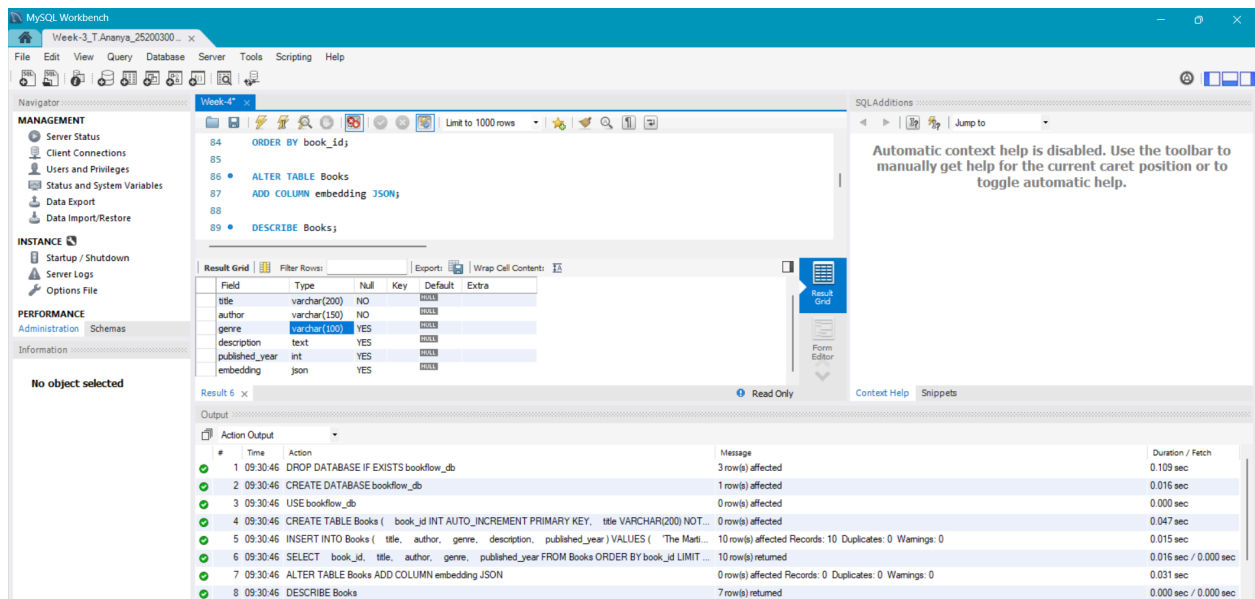
```
20     2011 ),
21
22     'Dune',
23     'Frank Herbert',
24     'Science Fiction',
25     'A story involving planets, politics, survival and space.',
26     1965 ),
27 (
28     'The Hitchhikers Guide to the Galaxy',
29     'Douglas Adams',
30     'Science Fiction',
31     'A humorous adventure across space and alien worlds.',
32     1979
33 ),
34 (
```

#	Time	Action	Message	Duration / Fetch
1	09:30:46	DROP DATABASE IF EXISTS bookflow_db	3 row(s) affected	0.109 sec
2	09:30:46	CREATE DATABASE bookflow_db	1 row(s) affected	0.016 sec
3	09:30:46	USE bookflow_db	0 row(s) affected	0.000 sec
4	09:30:46	CREATE TABLE Books (book_id INT AUTO_INCREMENT PRIMARY KEY, title VARCHAR(200) NOT NULL, author VARCHAR(100) NOT NULL, genre VARCHAR(50) NOT NULL, description VARCHAR(255) NOT NULL, published_year INT NOT NULL)	0 row(s) affected	0.047 sec
5	09:30:46	INSERT INTO Books (title, author, genre, description, published_year) VALUES ('The Marti...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0	0.015 sec

Query-4:



Query-5:



Query-6:

MySQL Workbench

Week-3_T_Ananya_25200300... x

File Edit View Query Database Server Tools Scripting Help

Navigator

MANAGEMENT

- Server Status
- Client Connections
- Users and Privileges
- Status and System Variables
- Data Export
- Data Import/Restore

INSTANCE

- Startup / Shutdown
- Server Logs
- Options File

PERFORMANCE

- Administration
- Schemas

No object selected

Week-4*

```

84 ORDER BY book_id;
85
86 ALTER TABLE Books
87 ADD COLUMN embedding JSON;
88
89 DESCRIBE Books;
90
91 UPDATE Books
92 SET embedding = JSON_ARRAY(0.90, 0.80, 0.20)
93 WHERE book_id = 1;
94
95 UPDATE Books
96 SET embedding = JSON_ARRAY(0.85, 0.75, 0.25)
97 WHERE book_id = 2;
98

```

SQLAdditions

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
1	09:30:46	DROP DATABASE IF EXISTS bookflow_db	3 row(s) affected	0.109 sec
2	09:30:46	CREATE DATABASE bookflow_db	1 row(s) affected	0.016 sec
3	09:30:46	USE bookflow_db	0 row(s) affected	0.000 sec
4	09:30:46	CREATE TABLE Books (book_id INT AUTO_INCREMENT PRIMARY KEY, title VARCHAR(200) NOT NULL, author VARCHAR(200) NOT NULL, genre VARCHAR(200) NOT NULL, description VARCHAR(200) NOT NULL, published_year YEAR(4) NOT NULL)	0 row(s) affected	0.047 sec
5	09:30:46	INSERT INTO Books (title, author, genre, description, published_year) VALUES ('The Mart...', '...', '...', '...', '...')	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0	0.015 sec
6	09:30:46	SELECT book_id, title, author, genre, published_year FROM Books ORDER BY book_id LIMIT 10	10 row(s) returned	0.016 sec / 0.000 sec
7	09:30:46	ALTER TABLE Books ADD COLUMN embedding JSON	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.031 sec
8	09:30:46	DESCRIBE Books	7 row(s) returned	0.000 sec / 0.000 sec
9	09:30:46	UPDATE Books SET embedding = JSON_ARRAY(0.90, 0.80, 0.20) WHERE book_id = 1	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.000 sec
10	09:30:46	UPDATE Books SET embedding = JSON_ARRAY(0.85, 0.75, 0.25) WHERE book_id = 2	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.000 sec

Query-7:

MySQL Workbench

Week-3_T_Ananya_25200300... x

File Edit View Query Database Server Tools Scripting Help

Navigator

MANAGEMENT

- Server Status
- Client Connections
- Users and Privileges
- Status and System Variables
- Data Export
- Data Import/Restore

INSTANCE

- Startup / Shutdown
- Server Logs
- Options File

PERFORMANCE

- Administration
- Schemas

No object selected

Week-4*

```

96 SET embedding = JSON_ARRAY(0.85, 0.75, 0.25)
97 WHERE book_id = 2;
98
99 UPDATE Books
100 SET embedding = JSON_ARRAY(0.80, 0.70, 0.30)
101 WHERE book_id = 3;
102
103 UPDATE Books
104 SET embedding = JSON_ARRAY(0.10, 0.20, 0.90)
105 WHERE book_id = 4;
106
107 UPDATE Books
108 SET embedding = JSON_ARRAY(0.15, 0.30, 0.80)
109 WHERE book_id = 5;
110

```

SQLAdditions

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
4	09:30:46	CREATE TABLE Books (book_id INT AUTO_INCREMENT PRIMARY KEY, title VARCHAR(200) NOT NULL, author VARCHAR(200) NOT NULL, genre VARCHAR(200) NOT NULL, description VARCHAR(200) NOT NULL, published_year YEAR(4) NOT NULL)	0 row(s) affected	0.047 sec
5	09:30:46	INSERT INTO Books (title, author, genre, description, published_year) VALUES ('The Mart...', '...', '...', '...', '...')	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0	0.015 sec
6	09:30:46	SELECT book_id, title, author, genre, published_year FROM Books ORDER BY book_id LIMIT 10	10 row(s) returned	0.016 sec / 0.000 sec
7	09:30:46	ALTER TABLE Books ADD COLUMN embedding JSON	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.031 sec
8	09:30:46	DESCRIBE Books	7 row(s) returned	0.000 sec / 0.000 sec
9	09:30:46	UPDATE Books SET embedding = JSON_ARRAY(0.90, 0.80, 0.20) WHERE book_id = 1	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.000 sec
10	09:30:46	UPDATE Books SET embedding = JSON_ARRAY(0.85, 0.75, 0.25) WHERE book_id = 2	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.000 sec
11	09:30:46	UPDATE Books SET embedding = JSON_ARRAY(0.80, 0.70, 0.30) WHERE book_id = 3	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.016 sec
12	09:30:46	UPDATE Books SET embedding = JSON_ARRAY(0.10, 0.20, 0.90) WHERE book_id = 4	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.000 sec
13	09:30:46	UPDATE Books SET embedding = JSON_ARRAY(0.15, 0.30, 0.80) WHERE book_id = 5	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.000 sec

Query-8:

MySQL Workbench

Week-3_T_Ananya_25200300...

File Edit View Query Database Server Tools Scripting Help

Navigator

MANAGEMENT

- Server Status
- Client Connections
- Users and Privileges
- Status and System Variables
- Data Export
- Data Import/Restore

INSTANCE

- Startup / Shutdown
- Server Logs
- Options File

PERFORMANCE

- Administration
- Schemas

No object selected

Week-4*

```

111 • UPDATE Books
112   SET embedding = JSON_ARRAY(0.95, 0.65, 0.20)
113   WHERE book_id = 6;
114
115 • UPDATE Books
116   SET embedding = JSON_ARRAY(0.30, 0.75, 0.90)
117   WHERE book_id = 7;
118
119 • UPDATE Books
120   SET embedding = JSON_ARRAY(0.88, 0.85, 0.15)
121   WHERE book_id = 8;
122
123 • UPDATE Books
124   SET embedding = JSON_ARRAY(0.20, 0.55, 0.95)
125   WHERE book_id = 9;

```

SQLAdditions

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
11	09:30:46	UPDATE Books SET embedding = JSON_ARRAY(0.80, 0.70, 0.30) WHERE book_id = 3	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.016 sec
12	09:30:46	UPDATE Books SET embedding = JSON_ARRAY(0.10, 0.20, 0.90) WHERE book_id = 4	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.000 sec
13	09:30:46	UPDATE Books SET embedding = JSON_ARRAY(0.15, 0.30, 0.80) WHERE book_id = 5	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.000 sec
14	09:30:46	UPDATE Books SET embedding = JSON_ARRAY(0.95, 0.65, 0.20) WHERE book_id = 6	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.000 sec
15	09:30:46	UPDATE Books SET embedding = JSON_ARRAY(0.30, 0.75, 0.90) WHERE book_id = 7	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.016 sec
16	09:30:46	UPDATE Books SET embedding = JSON_ARRAY(0.88, 0.85, 0.15) WHERE book_id = 8	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.015 sec
17	09:30:46	UPDATE Books SET embedding = JSON_ARRAY(0.20, 0.55, 0.95) WHERE book_id = 9	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.000 sec

Query-9:

MySQL Workbench

Week-3_T_Ananya_25200300...

File Edit View Query Database Server Tools Scripting Help

Navigator

MANAGEMENT

- Server Status
- Client Connections
- Users and Privileges
- Status and System Variables
- Data Export
- Data Import/Restore

INSTANCE

- Startup / Shutdown
- Server Logs
- Options File

PERFORMANCE

- Administration
- Schemas

No object selected

Week-4*

```

120   SET embedding = JSON_ARRAY(0.88, 0.85, 0.15)
121   WHERE book_id = 8;
122
123 • UPDATE Books
124   SET embedding = JSON_ARRAY(0.20, 0.55, 0.95)
125   WHERE book_id = 9;
126
127 • UPDATE Books
128   SET embedding = JSON_ARRAY(0.70, 0.90, 0.40)
129   WHERE book_id = 10;
130
131 • SELECT
132   book_id,
133   title,
134   embedding

```

SQLAdditions

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
11	09:30:46	UPDATE Books SET embedding = JSON_ARRAY(0.80, 0.70, 0.30) WHERE book_id = 3	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.016 sec
12	09:30:46	UPDATE Books SET embedding = JSON_ARRAY(0.10, 0.20, 0.90) WHERE book_id = 4	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.000 sec
13	09:30:46	UPDATE Books SET embedding = JSON_ARRAY(0.15, 0.30, 0.80) WHERE book_id = 5	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.000 sec
14	09:30:46	UPDATE Books SET embedding = JSON_ARRAY(0.95, 0.65, 0.20) WHERE book_id = 6	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.000 sec
15	09:30:46	UPDATE Books SET embedding = JSON_ARRAY(0.30, 0.75, 0.90) WHERE book_id = 7	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.016 sec
16	09:30:46	UPDATE Books SET embedding = JSON_ARRAY(0.88, 0.85, 0.15) WHERE book_id = 8	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.015 sec
17	09:30:46	UPDATE Books SET embedding = JSON_ARRAY(0.20, 0.55, 0.95) WHERE book_id = 9	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.000 sec
18	09:30:46	UPDATE Books SET embedding = JSON_ARRAY(0.70, 0.90, 0.40) WHERE book_id = 10	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.016 sec

Query-10:

MySQL Workbench

Week-3_T_Anyanya_25200300...

File Edit View Query Database Server Tools Scripting Help

Navigator

MANAGEMENT

- Server Status
- Client Connections
- Users and Privileges
- Status and System Variables
- Data Export
- Data Import/Restore

INSTANCE

- Startup / Shutdown
- Server Logs
- Options File

PERFORMANCE

- Administration
- Schemas

Information

No object selected

```

133 title,
134 embedding
135 FROM Books
136 ORDER BY book_id;
137
138 SELECT
139 book_id,
140 title,
141 author,
142 genre,
143 ROUND(
144     SQRT(
145         POW(
146             CAST(JSON_EXTRACT(embedding, '$[0]') AS DECIMAL(10,6)) - 0.90,
147             2

```

SQLAdditions

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
11	09:30:46	UPDATE Books SET embedding = JSON_ARRAY(0.80, 0.70, 0.30) WHERE book_id = 3	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.016 sec
12	09:30:46	UPDATE Books SET embedding = JSON_ARRAY(0.10, 0.20, 0.90) WHERE book_id = 4	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.000 sec
13	09:30:46	UPDATE Books SET embedding = JSON_ARRAY(0.15, 0.30, 0.80) WHERE book_id = 5	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.000 sec
14	09:30:46	UPDATE Books SET embedding = JSON_ARRAY(0.95, 0.65, 0.20) WHERE book_id = 6	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.000 sec
15	09:30:46	UPDATE Books SET embedding = JSON_ARRAY(0.30, 0.75, 0.90) WHERE book_id = 7	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.016 sec
16	09:30:46	UPDATE Books SET embedding = JSON_ARRAY(0.88, 0.85, 0.15) WHERE book_id = 8	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.015 sec
17	09:30:46	UPDATE Books SET embedding = JSON_ARRAY(0.20, 0.55, 0.95) WHERE book_id = 9	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.000 sec
18	09:30:46	UPDATE Books SET embedding = JSON_ARRAY(0.70, 0.90, 0.40) WHERE book_id = 10	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.016 sec
19	09:30:46	SELECT book_id, title, embedding FROM Books ORDER BY book_id LIMIT 0, 1000	10 row(s) returned	0.000 sec / 0.000 sec
20	09:30:47	SELECT book_id, title, author, genre, ROUND(SQRT(POW(CAST(USO...	3 row(s) returned	0.000 sec / 0.000 sec

Query-11:

MySQL Workbench

Week-3_T_Anyanya_25200300...

File Edit View Query Database Server Tools Scripting Help

Navigator

MANAGEMENT

- Server Status
- Client Connections
- Users and Privileges
- Status and System Variables
- Data Export
- Data Import/Restore

INSTANCE

- Startup / Shutdown
- Server Logs
- Options File

PERFORMANCE

- Administration
- Schemas

Information

No object selected

```

148 )
149 +
150 POW(
151     CAST(JSON_EXTRACT(embedding, '$[1]') AS DECIMAL(10,6)) - 0.80,
152     2
153 )
154 +
155 POW(
156     CAST(JSON_EXTRACT(embedding, '$[2]') AS DECIMAL(10,6)) - 0.20,
157     2
158 )
159 ),
160 4
161 ) AS l2_distance
162 FROM Books
163 WHERE embedding IS NOT NULL
164 ORDER BY l2_distance ASC

```

SQLAdditions

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
14	09:30:46	UPDATE Books SET embedding = JSON_ARRAY(0.95, 0.65, 0.20) WHERE book_id = 6	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.000 sec
15	09:30:46	UPDATE Books SET embedding = JSON_ARRAY(0.30, 0.75, 0.90) WHERE book_id = 7	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.016 sec
16	09:30:46	UPDATE Books SET embedding = JSON_ARRAY(0.88, 0.85, 0.15) WHERE book_id = 8	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.015 sec
17	09:30:46	UPDATE Books SET embedding = JSON_ARRAY(0.20, 0.55, 0.95) WHERE book_id = 9	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.000 sec
18	09:30:46	UPDATE Books SET embedding = JSON_ARRAY(0.70, 0.90, 0.40) WHERE book_id = 10	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.016 sec
19	09:30:46	SELECT book_id, title, embedding FROM Books ORDER BY book_id LIMIT 0, 1000	10 row(s) returned	0.000 sec / 0.000 sec
20	09:30:47	SELECT book_id, title, author, genre, ROUND(SQRT(POW(CAST(USO...	3 row(s) returned	0.000 sec / 0.000 sec

Query-12:

MySQL Workbench

Week-3_T_Ananya_25200300...

File Edit View Query Database Server Tools Scripting Help

Navigator

MANAGEMENT

- Server Status
- Client Connections
- Users and Privileges
- Status and System Variables
- Data Export
- Data Import/Restore

INSTANCE

- Startup / Shutdown
- Server Logs
- Options File

PERFORMANCE

- Administration
- Schemas

Information

No object selected

Week-4

Limit to 1000 rows

```

166
167 • SELECT
168     book_id,
169     title,
170     author,
171     genre,
172     ROUND(
173     (
174         CAST(JSON_EXTRACT(embedding, '$[0]') AS DECIMAL(10,6)) * 0.90
175         +
176         CAST(JSON_EXTRACT(embedding, '$[1]') AS DECIMAL(10,6)) * 0.80
177         +
178         CAST(JSON_EXTRACT(embedding, '$[2]') AS DECIMAL(10,6)) * 0.20
179     )
180 )
181
182     SORT(

```

Output

Action Output

#	Time	Action	Message	Duration / Fetch
14	09:30:46	UPDATE Books SET embedding = JSON_ARRAY(0.95, 0.65, 0.20) WHERE book_id = 6	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.000 sec
15	09:30:46	UPDATE Books SET embedding = JSON_ARRAY(0.30, 0.75, 0.90) WHERE book_id = 7	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.016 sec
16	09:30:46	UPDATE Books SET embedding = JSON_ARRAY(0.88, 0.85, 0.15) WHERE book_id = 8	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.015 sec
17	09:30:46	UPDATE Books SET embedding = JSON_ARRAY(0.20, 0.55, 0.95) WHERE book_id = 9	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.000 sec
18	09:30:46	UPDATE Books SET embedding = JSON_ARRAY(0.70, 0.90, 0.40) WHERE book_id = 10	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.016 sec
19	09:30:46	SELECT book_id, title, embedding FROM Books ORDER BY book_id LIMIT 0, 1000	10 row(s) returned	0.000 sec / 0.000 sec
20	09:30:47	SELECT book_id, title, author, genre, ROUND(SQRT(POW(CAST(JSON_EXTRACT(embedding, '\$[0]') AS DECIMAL(10,6)), 2	3 row(s) returned	0.000 sec / 0.000 sec
21	09:30:47	SELECT book_id, title, author, genre, ROUND((CAST(JSON_EXTRACT(embedding, '\$[0]') AS DECIMAL(10,6)) * 0.90	3 row(s) returned	0.000 sec / 0.000 sec

SQLAdditions

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Context Help Snippets

MySQL Workbench

Week-3_T_Ananya_25200300...

File Edit View Query Database Server Tools Scripting Help

Navigator

MANAGEMENT

- Server Status
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INSTANCE

- Startup / Shutdown
- Server Logs
- Options File

PERFORMANCE

- Administration
- Schemas

Information

No object selected

Week-4

Limit to 1000 rows

```

181
182 (
183     SQRT(
184         POW(
185             CAST(JSON_EXTRACT(embedding, '$[0]') AS DECIMAL(10,6)),
186             2
187         )
188         +
189         POW(
190             CAST(JSON_EXTRACT(embedding, '$[1]') AS DECIMAL(10,6)),
191             2
192         )
193         +
194         POW(
195             CAST(JSON_EXTRACT(embedding, '$[2]') AS DECIMAL(10,6)),
196             2
197         )
198     )
199 )

```

Output

Action Output

#	Time	Action	Message	Duration / Fetch
14	09:30:46	UPDATE Books SET embedding = JSON_ARRAY(0.95, 0.65, 0.20) WHERE book_id = 6	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.000 sec
15	09:30:46	UPDATE Books SET embedding = JSON_ARRAY(0.30, 0.75, 0.90) WHERE book_id = 7	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.016 sec
16	09:30:46	UPDATE Books SET embedding = JSON_ARRAY(0.88, 0.85, 0.15) WHERE book_id = 8	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.015 sec
17	09:30:46	UPDATE Books SET embedding = JSON_ARRAY(0.20, 0.55, 0.95) WHERE book_id = 9	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.000 sec
18	09:30:46	UPDATE Books SET embedding = JSON_ARRAY(0.70, 0.90, 0.40) WHERE book_id = 10	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.016 sec
19	09:30:46	SELECT book_id, title, embedding FROM Books ORDER BY book_id LIMIT 0, 1000	10 row(s) returned	0.000 sec / 0.000 sec
20	09:30:47	SELECT book_id, title, author, genre, ROUND(SQRT(POW(CAST(JSON_EXTRACT(embedding, '\$[0]') AS DECIMAL(10,6)), 2	3 row(s) returned	0.000 sec / 0.000 sec
21	09:30:47	SELECT book_id, title, author, genre, ROUND((CAST(JSON_EXTRACT(embedding, '\$[0]') AS DECIMAL(10,6)) * 0.90	3 row(s) returned	0.000 sec / 0.000 sec

SQLAdditions

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Context Help Snippets

MySQL Workbench

Week-3_T_Ananya_25200300...

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Information

No object selected

SQL Editor

```

194 CAST(JSON_EXTRACT(embedding, '$[2]') AS DECIMAL(10,6)),
195 2
196 )
197 )
198 *
199 SQRT(
200 POW(0.90, 2)
201 +
202 POW(0.80, 2)
203 +
204 POW(0.20, 2)
205 )
206 ),
207 4
208 ) AS cosine_similarity
209 FROM Books
210 WHERE embedding IS NOT NULL

```

SQL Additions

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
14	09:30:46	UPDATE Books SET embedding = JSON_ARRAY(0.95, 0.65, 0.20) WHERE book_id = 6	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.000 sec
15	09:30:46	UPDATE Books SET embedding = JSON_ARRAY(0.30, 0.75, 0.90) WHERE book_id = 7	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.016 sec
16	09:30:46	UPDATE Books SET embedding = JSON_ARRAY(0.88, 0.85, 0.15) WHERE book_id = 8	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.015 sec
17	09:30:46	UPDATE Books SET embedding = JSON_ARRAY(0.20, 0.55, 0.95) WHERE book_id = 9	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.000 sec
18	09:30:46	UPDATE Books SET embedding = JSON_ARRAY(0.70, 0.90, 0.40) WHERE book_id = 10	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.016 sec
19	09:30:46	SELECT book_id, title, embedding FROM Books ORDER BY book_id LIMIT 0, 1000	10 row(s) returned	0.000 sec / 0.000 sec
20	09:30:47	SELECT book_id, title, author, genre, ROUND(SQRT(POW(CAST(JSON_EXTRACT(embedding, '\$[0]') AS DECIMAL(10,6)), 2) + POW(CAST(JSON_EXTRACT(embedding, '\$[1]') AS DECIMAL(10,6)), 2) + POW(CAST(JSON_EXTRACT(embedding, '\$[2]') AS DECIMAL(10,6)), 2)) AS cosine_similarity FROM Books WHERE embedding IS NOT NULL	10 row(s) returned	0.000 sec / 0.000 sec

Output::

MySQL Workbench

Week-4*

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Navigator

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PERFORMANCE

- Administration
- Schemas

Information

No object selected

SQL Editor

```

207 4
208 ) AS cosine_similarity
209 FROM Books
210 WHERE embedding IS NOT NULL
211 ORDER BY cosine_similarity DESC

```

SQL Additions

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Context Help Snippets

Output

Result Grid

book_id	title	author	genre	published_year
1	The Martian	Andy Weir	Science Fiction	2011
2	Dune	Frank Herbert	Science Fiction	1965
3	The Hitchhikers Guide to the Galaxy	Douglas Adams	Science Fiction	1979
4	Pride and Prejudice	Jane Austen	Romance	1813
5	The Great Gatsby	F. Scott Fitzgerald	Classic	1925
6	Foundation	Isaac Asimov	Science Fiction	1951
7	The Hobbit	J.R.R. Tolkien	Fantasy	1937
8	Enders Game	Orson Scott Card	Science Fiction	1985
9	Harry Potter and the Sorcerers Stone	J.K. Rowling	Fantasy	1997
NULL	NULL	NULL	NULL	NULL

Books 12 × Result 13 Books 14 Result 15 Result 16

Week-3_T.Ananya_25200300... x

File Edit View Query Database Server Tools Scripting Help

SQL Editor

Week-4*

Limit to 1000 rows

GEMENT

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object selected

207

4

208

209

210

211

) AS cosine_similarity

FROM Books

WHERE embedding IS NOT NULL

ORDER BY cosine_similarity DESC

Result Grid

Filter Rows:

Export:

Wrap Cell Contents:

Field	Type	Null	Key	Default	Extra
book_id	int	NO	PRI	NULL	auto_increment
title	varchar(200)	NO		NULL	
author	varchar(150)	NO		NULL	
genre	varchar(100)	YES		NULL	
description	text	YES		NULL	
published_year	int	YES		NULL	
embedding	json	YES		NULL	

Books 12

Result 13 x

Books 14

Result 15

Result 16

Output

SQL workbench

Week-3_T.Ananya_25200300 ... x

Edit View Query Database Server Tools Scripting Help

ator..... Week-4* x

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ORMANCE

- ustration Schemas

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object selected

```
207      4
208      ) AS cosine_similarity
209 FROM Books
210 WHERE embedding IS NOT NULL
211 ORDER BY cosine_similarity DESC
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content: |

	book_id	title	embedding
▶	1	The Martian	[0.90, 0.80, 0.20]
	2	Dune	[0.85, 0.75, 0.25]
	3	The Hitchhikers Guide to the Galaxy	[0.80, 0.70, 0.30]
	4	Pride and Prejudice	[0.10, 0.20, 0.90]
	5	The Great Gatsby	[0.15, 0.30, 0.80]
	6	Foundation	[0.95, 0.65, 0.20]
	7	The Hobbit	[0.30, 0.75, 0.90]
	8	Enders Game	[0.88, 0.85, 0.15]
	9	Harry Potter and the Sorcerers Stone	[0.20, 0.55, 0.95]
*	NULL	NULL	NULL

Books 12 Result 13 Books 14 x Result 15 Result 16

Output

Action Output

#	Time	Action	Message
✓ 56	13:53:57	UPDATE Books SET embedding = JSON_ARRAY(0.15, 0.30, 0.80) WHERE book_id = 5	1 row

Week-3_T.Ananya_25200300... x

EditViewQueryDatabaseServerToolsScriptingHelp

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AGEMENT

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207
208
209
210
211

4

) AS cosine_similarity

FROM Books

WHERE embedding IS NOT NULL

ORDER BY cosine_similarity DESC

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

Fetch rows:

	book_id	title	author	genre	cosine_similarity
▶	1	The Martian	Andy Weir	Science Fiction	1
	2	Dune	Frank Herbert	Science Fiction	0.9986
	8	Enders Game	Orson Scott Card	Science Fiction	0.9983

ORMANCE

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object selected

Books 12

Result 13

Books 14

Result 15

Result 16 x

Output

Action Output

#	Time	Action	Message
50	15:51:57	UPDATE BOOK SET ...	1 row(s) affected Rows: 1